Introduction to Theoretical Computer Science

CS2490 - Session 22: Regular grammars and regular languages

Pierre Cagne

Fall 2026

October 9, 2026

Last time

A grammar G is given by:

- an alphabet Σ of terminal symbols
- an alphabet V of variables (or non-terminal symbols)
- a start variable $S \in V$
- a finite set P of production rules of the form $x \to y$ where $x, y \in (V \cup \Sigma)^*$ and x contains at least one variable

Last time

A grammar G is given by:

- an alphabet Σ of terminal symbols
- an alphabet V of variables (or non-terminal symbols)
- a start variable $S \in V$
- a finite set P of production rules of the form $x \to y$ where $x, y \in (V \cup \Sigma)^*$ and x contains at least one variable

G generates the language $\mathrm{L}(G)$ of all words derivable from S:

$$S \Rightarrow u_1 \Rightarrow u_2 \Rightarrow \cdots \Rightarrow w \in \Sigma^*$$

where each $\Rightarrow$ replaces an occurrence of the left-hand side of a production rule by its right-hand side.

Regular grammars

A grammar $G = (\Sigma, V, S, P)$ is regular if all production rules in P are of the form:

Regular grammars

A grammar $G = (\Sigma, V, S, P)$ is regular if all production rules in P are of the form:

- $A \to wB$, or

Regular grammars

A grammar $G = (\Sigma, V, S, P)$ is regular if all production rules in P are of the form:

- $A \to wB$, or
- $A \to w$

where $A, B \in V$ and $w \in \Sigma^*$.

Regular grammars

A grammar $G = (\Sigma, V, S, P)$ is regular if all production rules in P are of the form:

- $A \to wB$, or
- $A \to w$

where $A, B \in V$ and $w \in \Sigma^*$.

In each rule: the left-hand side is a single variable; and the right-hand side is a terminal word possibly followed by a single variable (not necessarily the same as the left-hand side).

Regular grammars

A grammar $G = (\Sigma, V, S, P)$ is regular if all production rules in P are of the form:

- $A \to wB$, or
- $A \to w$

where $A, B \in V$ and $w \in \Sigma^*$.

In each rule: the left-hand side is a single variable; and the right-hand side is a terminal word possibly followed by a single variable (not necessarily the same as the left-hand side).

(This is usually called a right-linear grammar. Left-linear grammars are defined similarly. A regular grammar is one that is either.)

Example

Consider the grammar $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

Example

Consider the grammar $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

A derivation of a word from S is necessarily of the form:

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

A derivation of a word from S is necessarily of the form:

S

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S$$

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S \Rightarrow 00S$$

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S \Rightarrow 00S \Rightarrow \cdots \Rightarrow 0^n S$$

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S \Rightarrow 00S \Rightarrow \cdots \Rightarrow 0^n S \Rightarrow 0^n 1A$$

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S \Rightarrow 00S \Rightarrow \cdots \Rightarrow 0^n S \Rightarrow 0^n 1A \Rightarrow 0^n 11A$$

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is $\mathrm{L}(G)$?

A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S \Rightarrow 00S \Rightarrow \cdots \Rightarrow 0^n S \Rightarrow 0^n 1A \Rightarrow 0^n 11A \Rightarrow \cdots \Rightarrow 0^n 11^m A$$

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is $\mathrm{L}(G)$?

A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S \Rightarrow 00S \Rightarrow \cdots \Rightarrow 0^n S \Rightarrow 0^n 1A \Rightarrow 0^n 11A \Rightarrow \cdots \Rightarrow 0^n 11^m A \Rightarrow 0^n 11^m$$

for some $n, m \geq 0$.

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is $\mathrm{L}(G)$?

A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S \Rightarrow 00S \Rightarrow \cdots \Rightarrow 0^n S \Rightarrow 0^n 1A \Rightarrow 0^n 11A \Rightarrow \cdots \Rightarrow 0^n 11^m A \Rightarrow 0^n 11^m$$

for some $n, m \geq 0$.That is, $\mathrm{L}(G) = \mathrm{L}(0^*11^*)$.

Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

Claim: $Xy \Rightarrow^* w$ if and only if w has parity X of as and parity y of bs. (Where E/e stands for even and O/o for odd.)

Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

Claim: $Xy \Rightarrow^* w$ if and only if w has parity X of as and parity y of bs. (Where E/e stands for even and O/o for odd.)

By induction on the length of the derivation of w. The only 1-step derivation is $Ee \Rightarrow \varepsilon$, and indeed ε has an even number of as and an even number of bs. Suppose the proposition true for derivations of length n. Any derivation of length $n+1$ is of the form $Xy \Rightarrow cX'y' \Rightarrow^* cw'$, where $X'y' \Rightarrow^* w'$ is a derivation of length n. By induction hypothesis, w' has parity X' of as and parity y' of bs. Case analysis on c and X', y' shows that the parity of cw' is as claimed.

Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

This is a regular grammar: all rules are of the form Variable $\to$ wordVariable or Variable $\to$ word.

Question: What is L(G)?

Claim: $Xy \Rightarrow^* w$ if and only if w has parity X of as and parity y of bs. (Where E/e stands for even and O/o for odd.)
By induction on the length of the derivation of w. The only 1-step derivation is $Ee \Rightarrow \varepsilon$, and indeed ε has an even number of as and an even number of bs. Suppose the proposition true for derivations of length n. Any derivation of length $n+1$ is of the form $Xy \Rightarrow cX'y' \Rightarrow^* cw'$, where $X'y' \Rightarrow^* w'$ is a derivation of length n. By induction hypothesis, w' has parity X' of as and parity y' of bs. Case analysis on c and X', y' shows that the parity of cw' is as claimed.

In particular, L(G) $= \{w \in \{a, b\}^* \mid w$ has an even number of as and an even number of bs$\}$.

From regular grammars to regular languages

Theorem

Regular grammars generate *regular languages*.

From regular grammars to regular languages

Theorem

Regular grammars generate *regular languages*.

Proof.

Let $G = (\Sigma, V, S, P)$ be a regular grammar.

From regular grammars to regular languages

Theorem

Regular grammars generate *regular languages*.

Proof.

Let $G = (\Sigma, V, S, P)$ be a regular grammar. Let's make an NFA $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ out of it!

From regular grammars to regular languages

Theorem

Regular grammars generate *regular languages*.

Proof.

Let $G = (\Sigma, V, S, P)$ be a regular grammar. Let's make an NFA $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ out of it!
We make parts of $\mathcal{A}$ for each rule in P, and then glue them together.

From regular grammars to regular languages

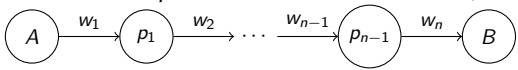
Theorem

Regular grammars generate *regular languages*.

Proof.

Let $G = (\Sigma, V, S, P)$ be a regular grammar. Let's make an NFA $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ out of it! We make parts of $\mathcal{A}$ for each rule in P, and then glue them together.

- For each rule p in P of the form $A \to wB$, craft the following fragment:



From regular grammars to regular languages

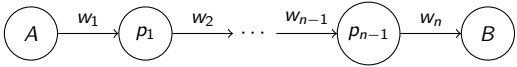
Theorem

Regular grammars generate *regular languages.*

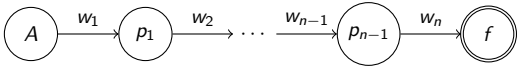
Proof.

Let $G = (\Sigma, V, S, P)$ be a regular grammar. Let's make an NFA $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ out of it! We make parts of $\mathcal{A}$ for each rule in P, and then glue them together.

- For each rule p in P of the form $A \to wB$, craft the following fragment:



- For each rule p in P of the form $A \to w$, craft the following fragment:



From regular grammars to regular languages

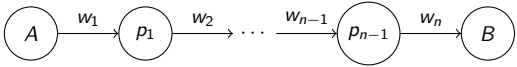
Theorem

Regular grammars generate *regular languages*.

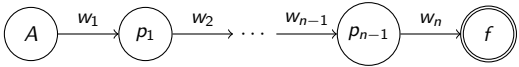
Proof.

Let $G = (\Sigma, V, S, P)$ be a regular grammar. Let's make an NFA $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ out of it! We make parts of $\mathcal{A}$ for each rule in P, and then glue them together.

- For each rule p in P of the form $A \to wB$, craft the following fragment:



- For each rule p in P of the form $A \to w$, craft the following fragment:



  If w is empty, just make A an accepting state.

From regular grammars to regular languages

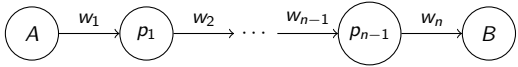
Theorem

Regular grammars generate *regular languages*.

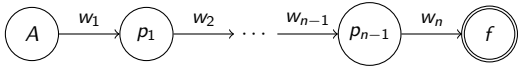
Proof.

Let $G = (\Sigma, V, S, P)$ be a regular grammar. Let's make an NFA $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ out of it! We make parts of $\mathcal{A}$ for each rule in P, and then glue them together.

- For each rule p in P of the form $A \to wB$, craft the following fragment:



- For each rule p in P of the form $A \to w$, craft the following fragment:



  If w is empty, just make A an accepting state.

Make S the only initial state. □

Example

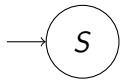
Apply to $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

Example

Apply to $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

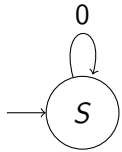
$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$



Example

Apply to $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

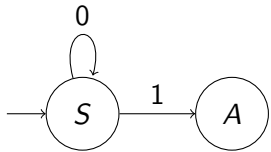
$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$



Example

Apply to $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

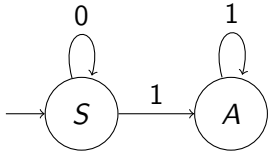
$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$



Example

Apply to $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

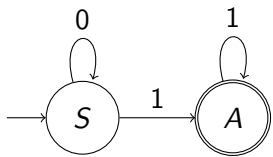
$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$



Example

Apply to $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$



Example

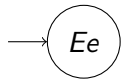
Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

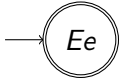
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

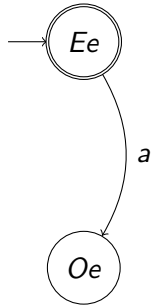
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

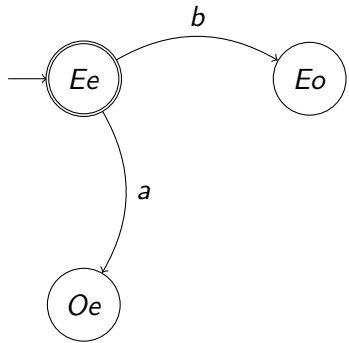
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

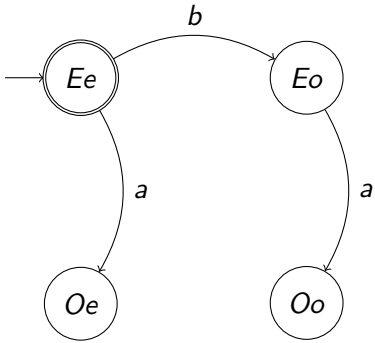
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

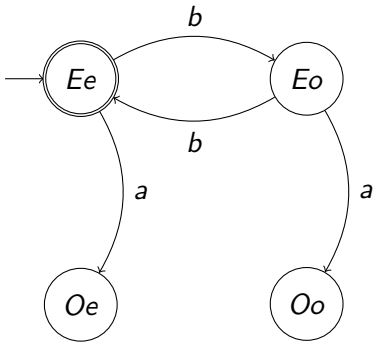
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

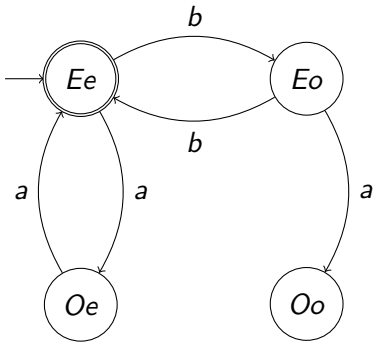
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

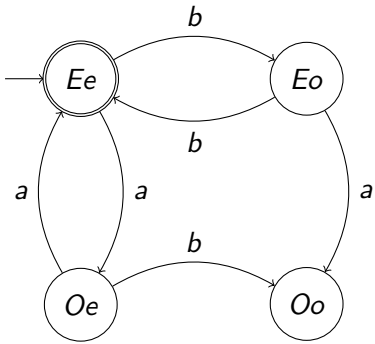
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

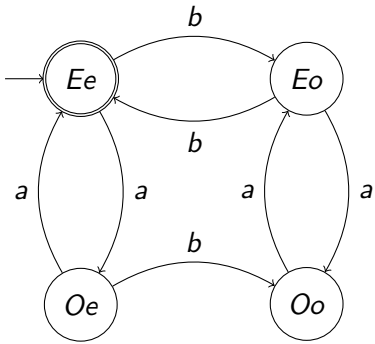
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

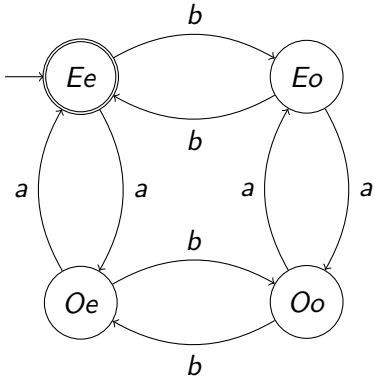
$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$



Example

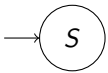
Consider the grammar $G = (\{a, b\}, \{S, A, B\}, S, P)$ where P contains the rules:

$$S \to abA, \quad A \to baB, \quad B \to aA \mid bb.$$

Example

Consider the grammar $G = (\{a, b\}, \{S, A, B\}, S, P)$ where P contains the rules:

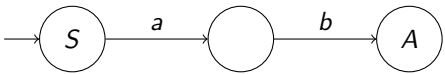
$$S \to abA, \quad A \to baB, \quad B \to aA \mid bb.$$



Example

Consider the grammar $G = (\{a, b\}, \{S, A, B\}, S, P)$ where P contains the rules:

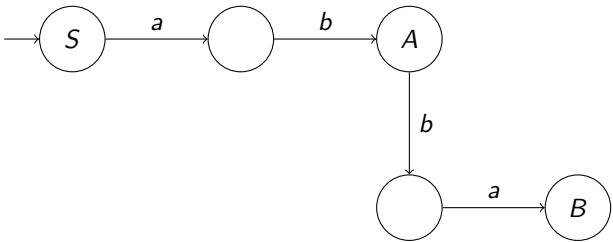
$$S \to abA, \quad A \to baB, \quad B \to aA \mid bb.$$



Example

Consider the grammar $G = (\{a, b\}, \{S, A, B\}, S, P)$ where P contains the rules:

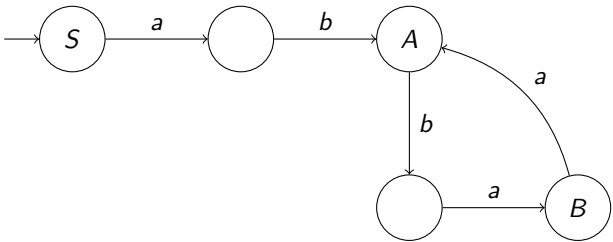
$$S \to abA, \quad A \to baB, \quad B \to aA \mid bb.$$



Example

Consider the grammar $G = (\{a, b\}, \{S, A, B\}, S, P)$ where P contains the rules:

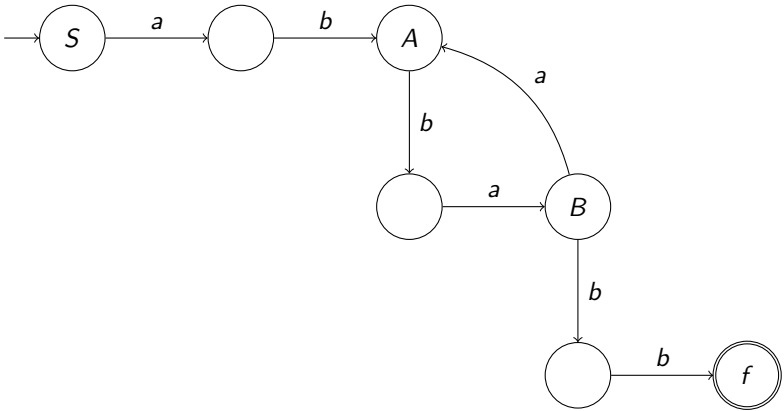
$$S \to abA, \quad A \to baB, \quad B \to aA \mid bb.$$



Example

Consider the grammar $G = (\{a, b\}, \{S, A, B\}, S, P)$ where P contains the rules:

$$S \to abA, \quad A \to baB, \quad B \to aA \mid bb.$$



From regular languages to regular grammars

Theorem

Regular languages are generated by regular grammars.

From regular languages to regular grammars

Theorem

Regular languages are generated by regular grammars.

Proof.

Let $\mathcal{A} = (\Sigma, Q, \delta, q_0, F)$ be a DFA.

From regular languages to regular grammars

Theorem

Regular languages are generated by regular grammars.

Proof.

Let $\mathcal{A} = (\Sigma, Q, \delta, q_0, F)$ be a DFA. Let's make a regular grammar $G = (\Sigma, V, S, P)$ out of it!

From regular languages to regular grammars

Theorem

Regular languages are generated by regular grammars.

Proof.

Let $\mathcal{A} = (\Sigma, Q, \delta, q_0, F)$ be a DFA. Let's make a regular grammar $G = (\Sigma, V, S, P)$ out of it!
The set of variables is Q itself.

From regular languages to regular grammars

Theorem

Regular languages are generated by regular grammars.

Proof.

Let $\mathcal{A} = (\Sigma, Q, \delta, q_0, F)$ be a DFA. Let's make a regular grammar $G = (\Sigma, V, S, P)$ out of it!
The set of variables is Q itself. The starting variable is the initial state q_0.

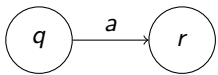
From regular languages to regular grammars

Theorem

Regular languages are generated by regular grammars.

Proof.

Let $\mathcal{A} = (\Sigma, Q, \delta, q_0, F)$ be a DFA. Let's make a regular grammar $G = (\Sigma, V, S, P)$ out of it! The set of variables is Q itself. The starting variable is the initial state q_0.

- For each transition 

, add the production rule $q \to ar$.

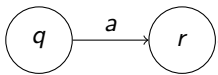
From regular languages to regular grammars

Theorem

Regular languages are generated by regular grammars.

Proof.

Let $\mathcal{A} = (\Sigma, Q, \delta, q_0, F)$ be a DFA. Let's make a regular grammar $G = (\Sigma, V, S, P)$ out of it! The set of variables is Q itself. The starting variable is the initial state q_0.

- For each transition 

, add the production rule $q \to ar$.
- For each final state 

, add the production rule $q \to \varepsilon$.

□

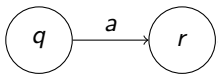
From regular languages to regular grammars

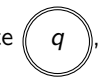
Theorem

Regular languages are generated by regular grammars.

Proof.

Let $\mathcal{A} = (\Sigma, Q, \delta, q_0, F)$ be a DFA. Let's make a regular grammar $G = (\Sigma, V, S, P)$ out of it! The set of variables is Q itself. The starting variable is the initial state q_0.

- For each transition 

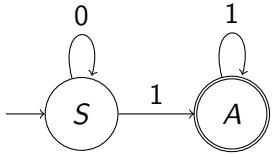
, add the production rule $q \to ar$.
- For each final state 

, add the production rule $q \to \varepsilon$.

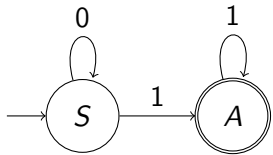
□

Note: the construction also works for NFAs, but you might have to add a new starting symbol S and rules $S \to i$ for any initial state i.

Example

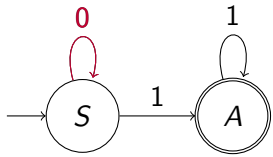


Example



This produces the grammar $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

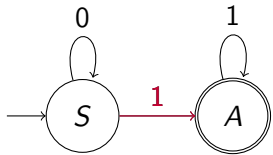
Example



This produces the grammar $G = (\{0, 1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S,$$

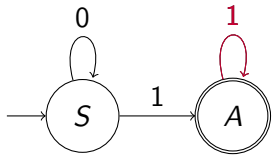
Example



This produces the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A,$$

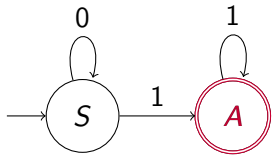
Example



This produces the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A,$$

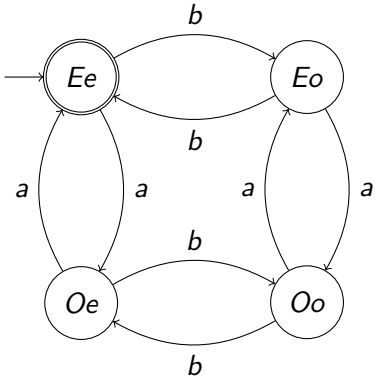
Example



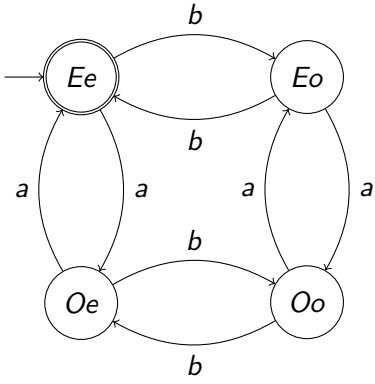
This produces the grammar $G = (\{0,1\}, \{S,A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

Example

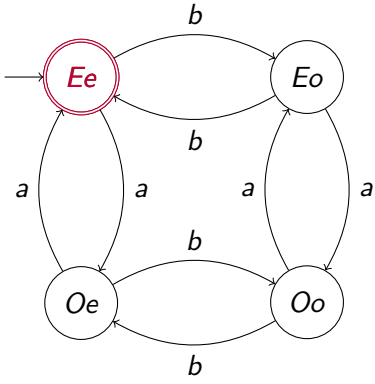


Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

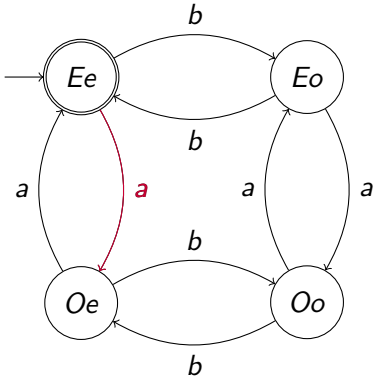
Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$Ee \to \varepsilon$

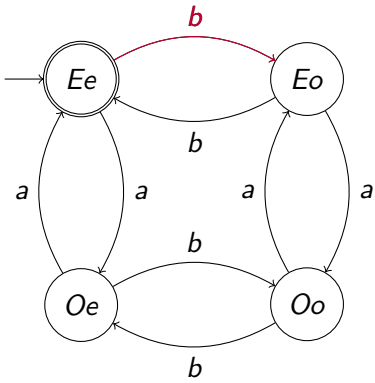
Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$Ee \rightarrow \varepsilon \mid aOe$

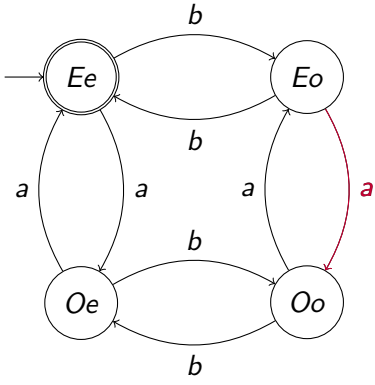
Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$Ee \to \varepsilon \mid aOe \mid bEo,$

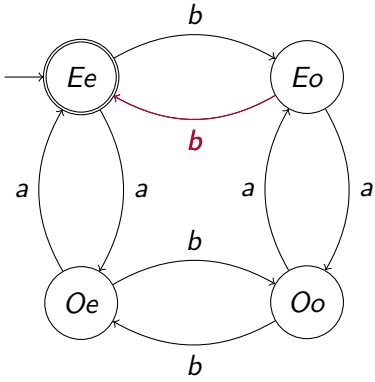
Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo$$

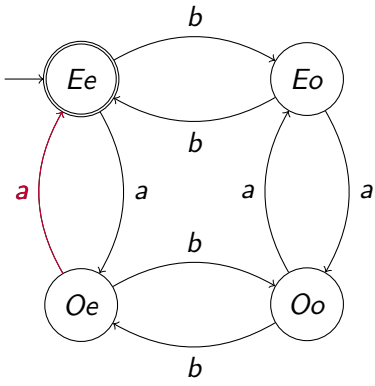
Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe,$$

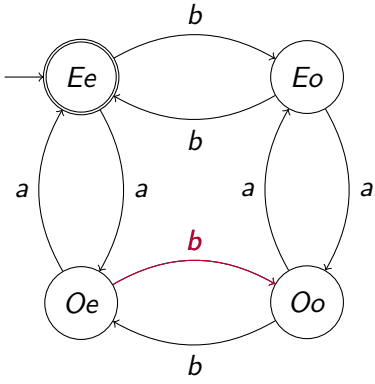
Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe$$

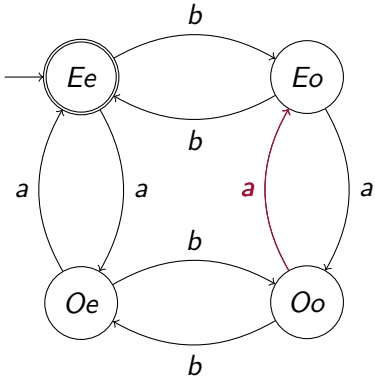
Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo,$$

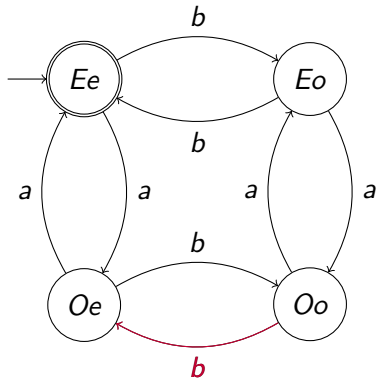
Example



This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo$$

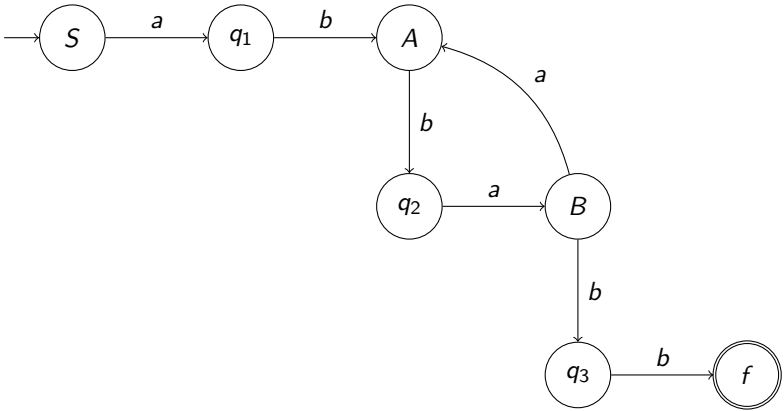
Example



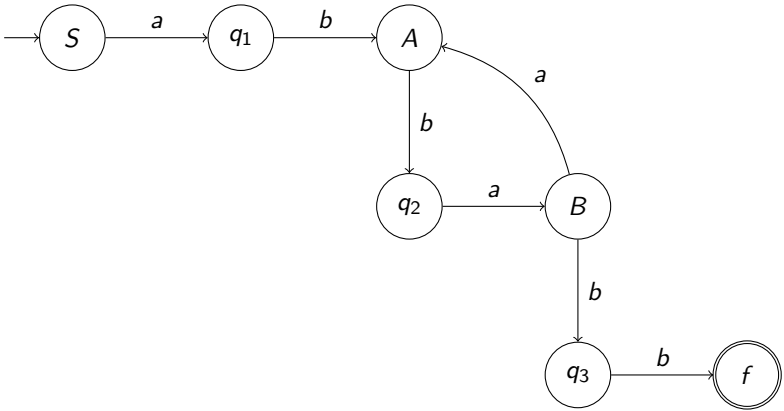
This produces the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

Example

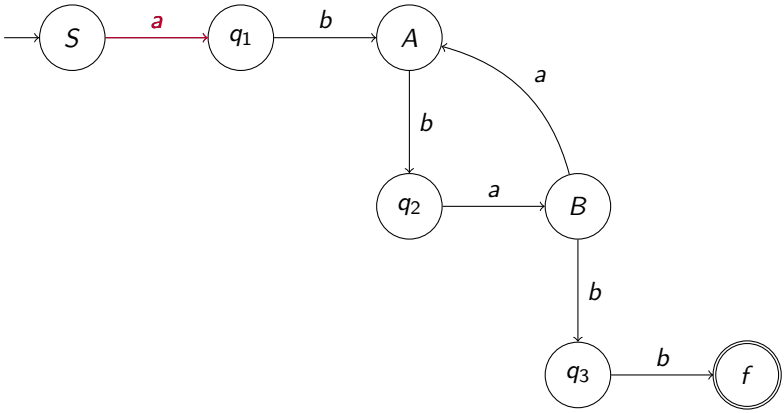


Example



This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

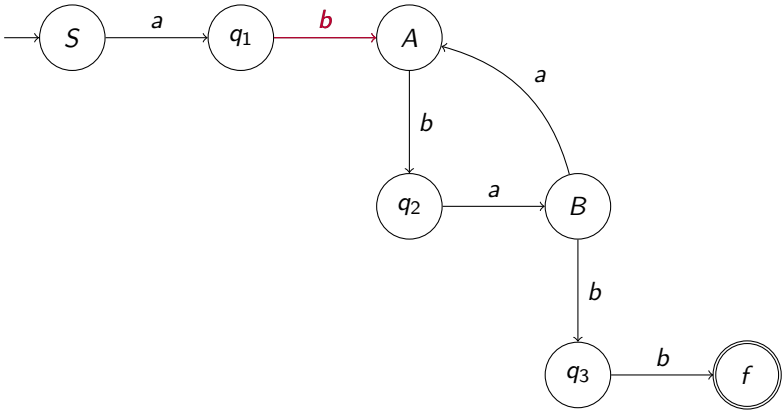
Example



This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

$S \to aq_1$,

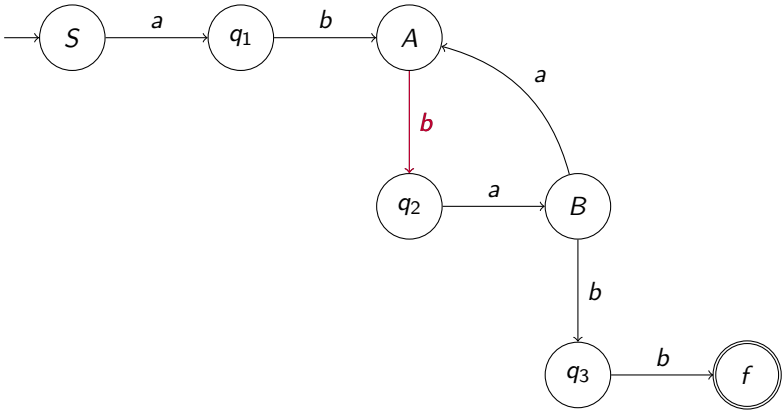
Example



This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

$S \to aq_1, \quad q_1 \to bA,$

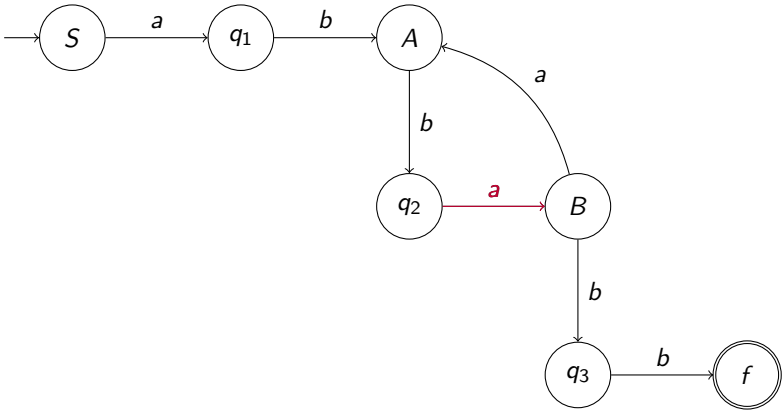
Example



This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

$S \to aq_1, \quad q_1 \to bA, \quad A \to bq_2,$

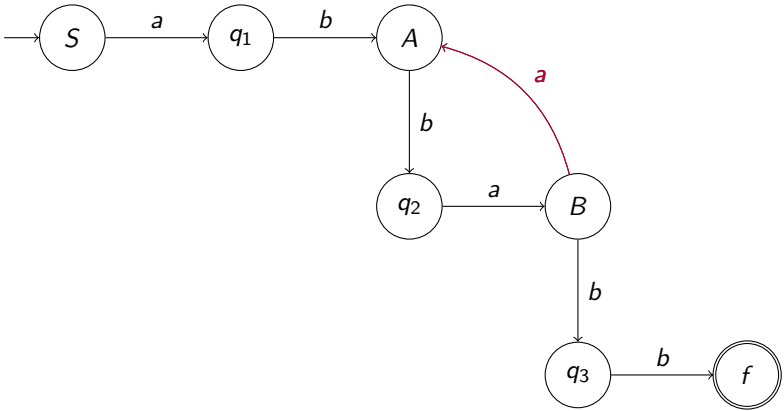
Example



This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

$S \to aq_1, \quad q_1 \to bA, \quad A \to bq_2, \quad q_2 \to aB,$

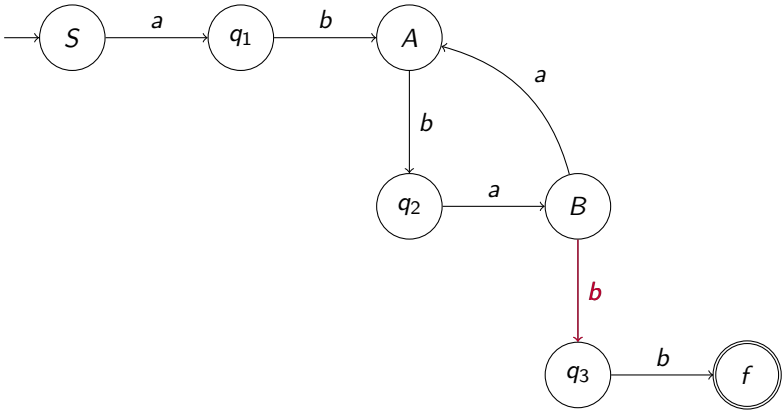
Example



This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

$S \to aq_1, \quad q_1 \to bA, \quad A \to bq_2, \quad q_2 \to aB, \quad B \to aA,$

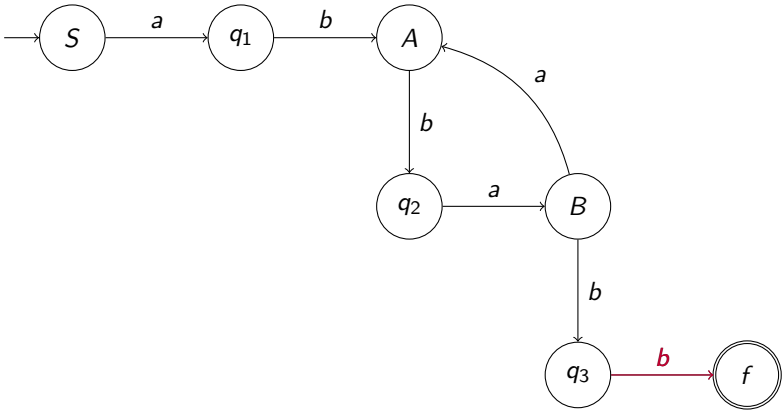
Example



This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

$S \to aq_1, \quad q_1 \to bA, \quad A \to bq_2, \quad q_2 \to aB, \quad B \to aA, \quad B \to bq_3,$

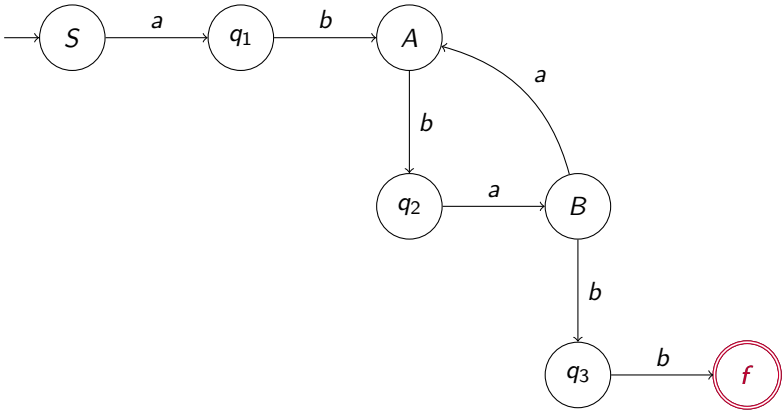
Example



This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

$S \to aq_1, \quad q_1 \to bA, \quad A \to bq_2, \quad q_2 \to aB, \quad B \to aA, \quad B \to bq_3, \quad q_3 \to bf,$

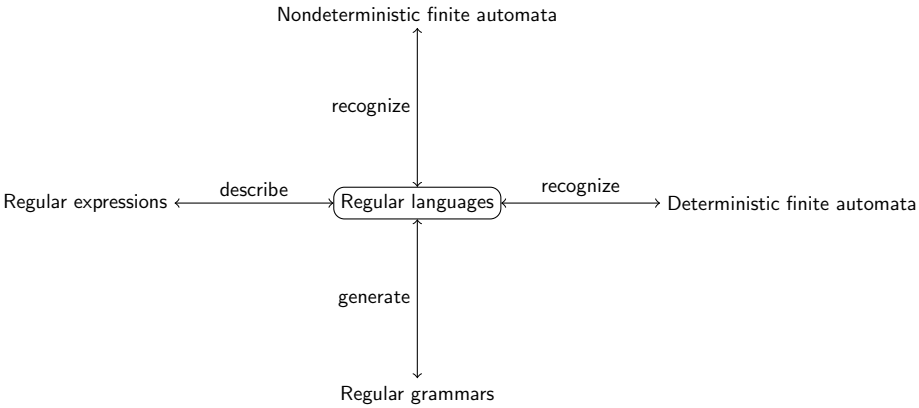
Example



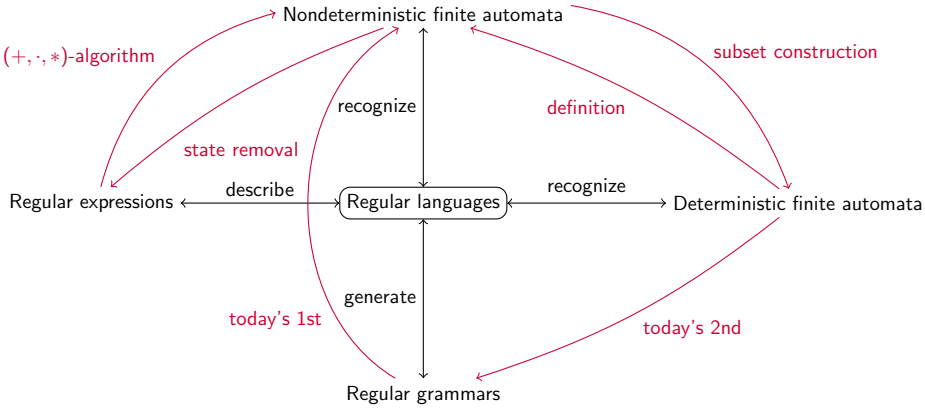
This produces the grammar $G = (\{a, b\}, \{S, A, B, q_1, q_2, q_3, f\}, S, P)$ where P contains the rules:

$S \to aq_1, \quad q_1 \to bA, \quad A \to bq_2, \quad q_2 \to aB, \quad B \to aA, \quad B \to bq_3, \quad q_3 \to bf, \quad f \to \varepsilon$

Summary


Nondeterministic finite automata
recognize
Regular expressions
describe
Regular languages
recognize
Deterministic finite automata
generate
Regular grammars

Summary


Nondeterministic finite automata
$(+, \cdot, *)$-algorithm
subset construction
recognize
definition
state removal
Regular expressions
describe
Regular languages
recognize
Deterministic finite automata
generate
today's 1st
today's 2nd
Regular grammars